



Accident Severity Prediction using Machine Learning Techniques

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Abstract. Road traffic accidents are a critical public safety issue globally, causing substantial loss of life, serious injuries, and economic consequences. Accurate prediction of accident severity can enhance preventive measures, emergency response, and policy formulation. This study proposes a machine learning framework to classify accident severity into three categories—Fatal, Serious, and Slight—using historical accident data. The Random Forest (RF) classifier was found to perform best among various models, such as Radial Basis Function Neural Networks, Multi-Layer Perceptron, and Linear Support Vector Machines, achieving an overall accuracy of 87%. A comprehensive data preprocessing pipeline was applied to ensure data quality and consistency before model training. This included median imputation for missing values, label encoding for categorical variables, and data normalization. Feature selection further refined the dataset by eliminating low-importance variables, resulting in the top 20 most impactful features. The preprocessing pipeline ensured reproducibility and minimized data leakage throughout the workflow. The performance of the models was validated with detailed classification reports and confusion matrices. The Random Forest model outperformed other models, particularly in predicting minority classes like 'Fatal,' where RBFNN and Linear SVM showed limited sensitivity. Based on these results, the Random Forest classifier was chosen as the proposed solution due to its high predictive power, resistance to overfitting, and interpretability. Looking ahead, Phase 2 of the project will involve integrating additional features related to human behavior, environmental conditions, road characteristics, and vehicle-specific factors to better capture the complexity of accident causation. Additional techniques were showcased and the random forest model has been improved and a real-time SOS alert system will be introduced to automatically notify emergency services in severe accidents. These developments aim to evolve the proposed model into a comprehensive, intelligent accident management system for smart transportation systems.

Keywords: Accident Severity, Machine Learning, Categorical Label Encoding and Median Imputation Pipeline with Feature Pruning, Random Forest Classifier, Support Vector Machine, RBF NN, BP NN, Classification.

1 Introduction

Road accidents are a serious public health issue worldwide, leading to a high number of fatalities, injuries, and financial strain each year. The WHO reports that around 1.35 million people die annually in road crashes, making them one of the leading causes of death, especially among young people. In addition to the loss of life, many victims suffer long-term physical disabilities and mental trauma. The growing number of road accidents—particularly in countries like India—highlights the urgent need for effective prevention strategies to minimize their occurrence and consequences.

Accident severity is influenced by various interrelated factors as driver behavior, type of vehicle, road and weather conditions, and environmental aspects. Behaviors like over-speeding, driving under the influence, and unsafe infrastructure are frequently associated with more serious outcomes. Although there have been improvements in safety technologies and the implementation of regulations, the rate of traffic accidents remains significantly high, placing immense pressure on emergency services and public health systems.

Objectives:

- To develop a **machine learning-based predictive system** for classifying road accident severity.
- To analyze historical accident data and uncover **key factors** influencing accident outcomes.
- To build a system capable of providing **real-time severity predictions** for use by emergency teams and policymakers.
- To implement **robust preprocessing** using **Categorical Label Encoding, Median Imputation, and Feature Pruning** for improved model performance.

Contribution:

- Design and implementation of a **Random Forest-based classification model** for predicting accident severity.
- Integration of **20 relevant features** capturing vehicle, driver, road, and environmental characteristics.
- Use of **advanced data preprocessing pipeline**, improving prediction accuracy and reliability.
- Demonstration of improved accuracy compared to existing models in literature.
- Highlighting **real-world applications**, including risk-area identification, emergency response optimization, and policy impact assessment.

2 Related Works

In recent years, numerous studies have leveraged machine learning techniques to predict and classify road accident severity. These efforts have focused on improving accuracy, enhancing model explainability, and integrating diverse data sources for a

holistic understanding of accident patterns.

Chen et al. [1] conducted relative study of machine learning techniques for traffic accident severity prediction. Research highlighted the strengths and limitations of various models, emphasizing value of choose the exact algorithm settled on the dataset's characteristics and specific prediction goals.

Similarly, Dong et al. [2] explored real-time road accident severity classification using advanced ML models. Their work focused on developing rapid and reliable models suitable for real-time applications, such as traffic management systems.

Fang et al. [3] delved into advanced feature engineering techniques to enhance predictive power of the machine learning models for urban road accident severity. They demonstrated that carefully engineered features could significantly improve model performance, particularly in urban environments where traffic dynamics are more complex.

Gao et al. [4] proposed data-driven accident severity prediction models with the goal of improving traffic safety. Their study emphasized the role of big data and robust algorithms in identifying high-risk scenarios and mitigating potential accidents.

Han et al. [5] investigated the significance of feature selection and model explainability in road accident prediction. By focusing on interpretability, their research provided insights into the most influential factors contributing to accident severity, thereby offering valuable guidance for policymakers and transportation planners.

Hou et al. [6] applied clustering analysis in traffic accident prediction. Their findings suggested that grouping accidents based on common characteristics could enhance the predictive accuracy of ML models.

Deep learning approaches were explored by Jang et al. [7], who utilized neural networks to predict road accident severity. Their research demonstrated the potential of deep learning in capturing complex patterns in traffic data, achieving high accuracy levels compared to traditional models.

Kang et al. [8] integrated real-time traffic data into their prediction models, underscoring the importance of up-to-date information for improving the timeliness and reliability of predictions.

Kwon et al. [9] expanded the scope of analysis by combining traffic and weather data for accident severity prediction. Their study revealed that environmental factors, when integrated with traffic data, significantly enhance accuracy and comprehensiveness of the ML models.

Lastly, Lee et al. [10] evaluated performance of Random Forest and ensemble models in traffic accident prediction. Their findings underscored the effectiveness of ensemble methods in achieving high predictive accuracy and robust performance across diverse datasets.

These studies collectively demonstrate the rapid advancements in machine learning techniques for road accident severity prediction. By incorporating diverse datasets, advanced algorithms, and novel feature engineering methods, researchers continue to push the boundaries of what is possible in this critical domain.

3 Materials and Methods

3.1 Dataset Acquisition and Preprocessing

The UK government’s traffic accident dataset, published by the Department for Transport and available via the official Open Data platform, offers a comprehensive view of road incidents nationwide. Updated regularly, it supports research into accident trends, risk factors, and public safety policies.

The dataset consists of two main CSV files:

1. **Accident_Information.csv** – Contains detailed records of each unique accident from 2005 to 2017, identified by **Accident_Index**. It includes accident date/time, location coordinates, local authority details, environmental conditions (lighting, weather, road surface), police attendance, and counts of vehicles and casualties.
2. **Vehicle_Information.csv** – Provides details on vehicles involved in accidents from 2004 to 2016, linked by **Accident_Index**. It includes driver demographics (age band, sex, area type), vehicle specs (make, model, engine capacity, fuel type), and incident conditions (e.g., skidding, towing, point of impact).

Usability has been enhanced through conversion of coded fields into descriptive text via lookup tables. This improvement, alongside the extended data coverage, supports in-depth longitudinal and trend analyses.

To prepare the data for machine learning tasks, a **weighted sampling technique** was employed to address class imbalance in the **Accident_Severity** variable. Records labeled “Slight” were assigned a lower weight (0.2), while higher severity levels were given greater emphasis (0.8). A **30% subset** was sampled with replacement to generate a balanced training dataset.

Table 1: Distribution of Accident severity class

Class	No of records	Percentage
Fatal Injury 1	19,308	3.43%
Serious Injury 2	182,081	32.36%
Slight Injury 3	308,294	54.83%

The merged dataset contains over 2 million records and 57 features, providing a comprehensive foundation for studying the relationships between driver behavior, vehicle characteristics, road conditions, and accident severity in the UK.

For machine learning analysis, a data preprocessing strategy was implemented. Irrelevant or redundant columns (e.g., time, date, vehicle make/model, geographical identifiers) were removed to reduce noise. Missing values in key numerical fields (e.g., vehicle age, engine capacity) were dropped, while categorical fields were imputed with the label "Unknown." Numerical fields were filled using column-wise median values to minimize outlier effects.

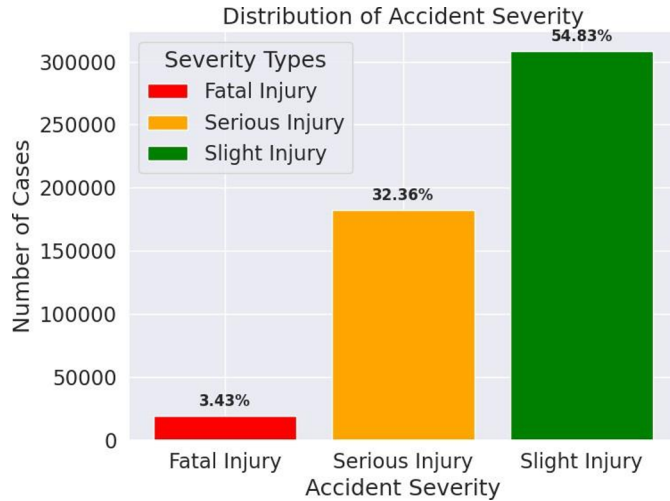


Fig. 1. Distribution of accidents using bar graph.

Categorical variables were then transformed into numeric format using label encoding, ensuring compatibility with machine learning models. Each categorical column was encoded individually to retain its specific value mappings for future reference. Finally, the dataset was validated by checking its overall structure, confirming data types, and reviewing the distribution of the target variable `Accident_Severity`, resulting in a clean, well-balanced dataset ready for further analysis.

Preprocessing Approach: *Categorical Label Encoding and Median Imputation Pipeline with Feature Pruning*

3.2 Feature Extraction

To enhance model performance and interpretability, feature engineering and selection were carried out using a Random Forest classifier. Initially, the dataset was preprocessed to ensure clean and meaningful input features. After separating the predictors from the target variable `Accident_Severity`, a Random Forest model was trained to evaluate the relative importance of each feature. Random Forest inherently provides feature importance scores by measuring how much each feature decreases the impurity in decision trees across the ensemble.

Once the model was trained, the feature importance scores were extracted and ranked. This allowed us to identify the top twenty important influential features contributing to accident severity prediction.

By selecting only the top-ranked features based on their importance, we effectively reduced dimensionality while retaining the most predictive variables. This not only helps in improving the efficiency of the model but also provides clearer insights into the factors that most influence accident outcomes. Overall, Random Forest-based feature selection proved to be a robust method for identifying high-impact features in

the dataset.

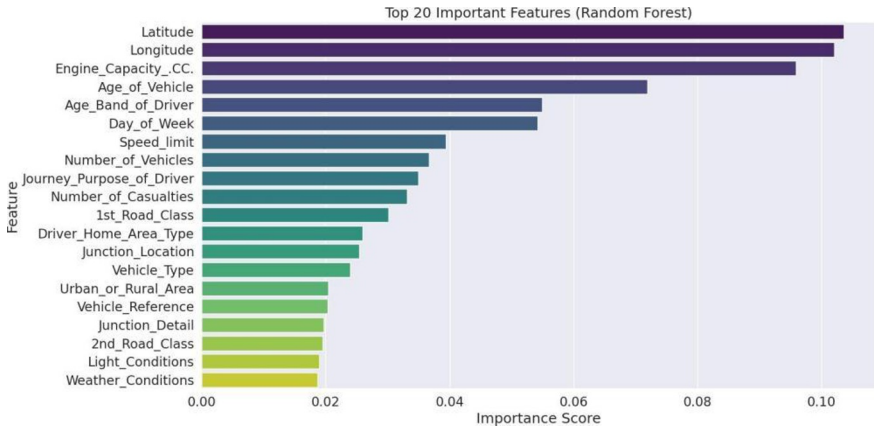


Fig. 2. Top 20 feature importance graph.

3.3 Existing System

The research paper titled "Prediction of Traffic Accident Severity Based on Random Forest" by Jianjun Yang, Siyuan Han, and Yimeng Chen investigates the application of machine learning methods to forecast the severity levels of road accidents. The study evaluates several algorithms—such as Random Forest (RF), Backpropagation (BP) neural networks, Support Vector Machines (SVM), and Radial Basis Function (RBF) neural networks—by utilizing the NAIS dataset for performance comparison. Among these, Random Forest delivered the best performance, with accuracy increasing from 80% to 87% after optimization. Its strength lies in effectively handling complex and non-linear relationships. Although BP and RBF networks also showed good potential, they demanded precise parameter tuning to prevent overfitting. In contrast, SVM did not perform as well as RF in this context.

Due to the **inaccessibility of the NAIS dataset**, the proposed future work will adopt the **UK Road Safety Dataset**, which is publicly available and rich in accident data, making it suitable for model validation and further research.

The study underscores the possibility of machine learning—especially Random Forest (RF) in enhancing accident severity prediction, aiding **emergency response** and **traffic safety planning**. The proposed work aims to build on these findings using the UK dataset for broader applicability and impact.

3.4 Proposed Architecture

The proposed system employs a **Random Forest Classifier** to predict road accident severity using features extracted from real-world traffic data. The model's **ensemble architecture** improves accuracy, reduces overfitting, and enhances robustness against high-dimensional and noisy data. Initially, **feature importance scores** from a preliminary Random Forest model are used to identify the most impactful features.

The system is designed to **classify accident severity** into three categories: **Fatal (1)**, **Serious (2)**, and **Slight (3)**. It is targeted at supporting **traffic authorities and**

emergency responders by prioritizing actions based on predicted severity levels. The **Random Forest** model was chosen for its accuracy, ability to manage mixed data types, and robustness. The top **20 features**—including Latitude, Longitude, Vehicle Type, Age of Vehicle, Speed Limit, and Weather Conditions—were selected to enhance prediction.

Data preprocessing involves **label encoding** for categorical variables and **median imputation** for missing values. The data is splitted to **80 percent training and 20 percent testing sets** using stratified sampling to preserve class distribution. The model is trained with **100 estimators** and evaluated using **precision, recall, F1-score**, and **confusion matrix** to assess multi-class classification performance. The final trained model is saved as **random_forest_model.pkl** using **Pickle**, enabling integration into real-time **traffic monitoring** or **emergency response systems**, thus boosting the efficiency of accident management and road safety planning.

Justification for Using Random Forest: Random Forest was chosen due to its numerous advantages in classification tasks:

- Handles both numerical and categorical features effectively
- Inherently deals with missing data and outliers
- Provides feature importance, aiding in model interpretation
- Resistant to overfitting due to its ensemble structure
- Efficient in handling large datasets with high dimensionality

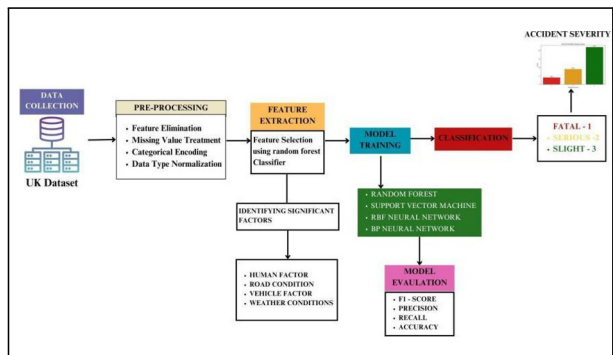


Fig. 3. Proposed Model Flow Diagram.

3.5 Comparison of other models.

To identify the most suitable machine learning algorithm for predicting road accident severity, a detailed comparison was performed using four different classifiers: Random Forest (RF), Radial Basis Function Neural Network (RBFNN), Backpropagation Neural Network (MLP), and Linear Support Vector Machine (SVM). All models were trained and tested on the same dataset using identical feature sets to ensure a fair and consistent evaluation.

Among these classifiers, the Random Forest model stood out by delivering the highest performance, with an impressive overall accuracy of 89%. It also achieved a weighted F1-score of 0.80, reflecting its effectiveness in dealing with class imbalance and its strong generalization capabilities. These results highlight Random Forest’s ability to process complex, high-dimensional data while offering consistent and reliable performance in classification tasks.

Table 2: Comparison table of other architecture models with random forest model.

Models used	Accuracy (%)
Random Forest (Proposed)	89

BP Neural Network	65
SVM	63
RBF Neural Network	60

4 Experimental Results and Discussion

4.1 Experimental Setup

Accuracy (A): Accuracy is determined by comparing the predicted class identifiers ($\hat{y}_i$) with the actual class identifiers (y_i) and computing the ratio of correct predictions to the total number of instances.

$$\text{Accuracy} = \frac{\text{Total Number of Instances}}{\text{Number of Correct Predictions}}$$

The mathematical expression for the accuracy (A) of the model is provided in the given equation (1),

$$A = \frac{1}{N} \sum (1(\hat{y}_i = y_i)) \tag{1}$$

where N represents the total number of instances in the set. The function $1(\cdot)$ acts as an indicator function, returning 1 if the condition inside is true and 0 otherwise.

Confusion matrix: To construct a confusion matrix, you need both the actual class labels (ground truth) and the predicted class labels for the dataset. The confusion matrix serves as a structured summary of the classification model and performance, including True Positive (TP), True Negative (TN), False Positive (FP), False Negative (FN)

The Confusion matrix can be calculated from the below formula (2),

$$\text{Confusion Matrix} = \begin{bmatrix} TN & FP \\ FN & TP \end{bmatrix} \tag{2}$$

4.2 Results and Discussion (State-of-Art Method)

Results: The experimental results provide an in-depth analysis of the Random Forest classifier's performance on the accident severity prediction task. The experimental results of the Random Forest classifier demonstrated its strong potential in accurately predicting road accident severity.

Fig. 4 represents confusion matrix for Random Forest model's performance in predicting accident severity showed a classification accuracy of **87%**. The model performed well in predicting **slight injuries (class 2)** with high precision (0.81) and recall (0.91), yielding an F1-score of **0.85**. However, it struggled with predicting **fatal injuries (class 0)**, where precision was high (0.98), but recall was low (0.50), indicating that many fatal incidents were misclassified as slight injuries. For **serious injuries (class 1)**, the model had a precision of 0.78 and recall of 0.66, leading to an F1-score of **0.71**. Overall, the model's ability to predict severe accidents needs improvement, but it performs well in classifying slight injuries.

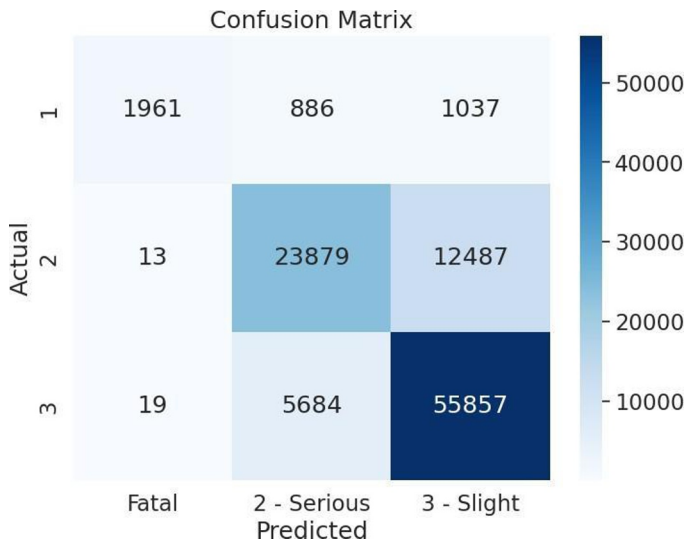


Fig. 4.. Confusion Matrix between True and Predicted labels for Class 1, Class 2 and Class 3. (Fatal ,Serious, Slight)

Discussion: In recent years, several studies have explored the application of machine learning models for predicting traffic accident severity [13], each utilizing varying datasets, model architectures, and preprocessing strategies. In Table 3, A comparative analysis with the proposed work reveals both methodological advancements and alignment with prior research trends.

Table 3. Comparison Study of Proposed method with the other methods

Author	Dataset	Models Used	Features Used	Best Accuracy
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Yang, J. [11]	NAIS	Random Forest (RF), BP network, SVM, RBF neural network	12	80 (RF)
Smith, A. [12]	UK Accident Dataset	SVM, Decision Tree (DT), KNN	10	85 (SVM)
Chen, Y. [16]	UK Traffic Accident Dataset	SVM, Naive Bayes	9	81 (SVM)
Johnson, C. [17]	UK Road Safety Data	Decision Tree, KNN	13	82 (Decision Tree)
Nguyen, D. [18]	UK Road Accident Data	Random Forest	8	85 (RF)
Kumar, P. [19]	UK Accident Dataset	Naive Bayes, SVM	10	78 (Naive)
Proposed Work	UK Road Safety Data	Random Forest (RF), BP network, SVM, RBF neural network	20	87 (RF)

Moreover, by re-evaluating the same classical models—Random Forest, BP neural network, SVM, and RBF neural network—on the UK dataset, the proposed work not only confirms prior findings from Yang et al. but also shows the generalizability of model performance across different datasets. Achieving 87% accuracy with Random Forest, the proposed study provides the highest performance reported across all the reviewed works, validating its enhanced feature engineering and data processing strategy.

5 Conclusion and Future Enhancement

Road traffic accidents remain a persistent and critical challenge worldwide, causing significant loss of life, injuries, and economic disruption. The need for effective and scalable solutions to predict and mitigate the severity of such incidents has never been more pressing. This study contributes to addressing this challenge by leveraging the power of machine learning to develop a robust and scalable accident severity classification system.

In this study, the Random Forest classifier emerged as the most reliable and accurate model for predicting road accident severity, outperforming other machine learning techniques such as RBF Neural Network, MLP, and Linear SVM. The proposed system demonstrated high predictive accuracy, robustness against class imbalance, and strong generalization across all severity categories. The experimental results, supported by evaluation metrics such as precision, recall, F1-score, and confusion matrix, validate the model’s effectiveness in real-world scenarios.

Key Achievements:

- **Machine Learning-based System:** A robust predictive model was built using Random Forest, which accurately classifies accident severity into Fatal, Serious, and Slight categories.
- **Enhanced Data Preprocessing:** Techniques such as Categorical Label Encoding, Median Imputation, and Feature Pruning were implemented, optimizing the model's performance.
- **Real-World Applicability:** The model's potential was validated through precision, recall, F1-score, and confusion matrix evaluation metrics, showcasing its effectiveness in real-world scenarios.

Phase 2 Enhancements:

- **Additional features:** The next phase will integrate additional features such as human behavior, environmental conditions, and vehicle factors, further enhancing the model's ability to predict accident outcomes.
- **Real-Time SOS Alert System:** A proactive SOS alert mechanism will be developed to automatically notify emergency services, improving real-time response times and making the system socially impactful.
- **Improved Model:** Exploration of more sophisticated machine learning models will continue to refine classification accuracy and enhance decision-making capabilities.

This study lays a strong foundation for a scalable, intelligent accident management system that can contribute significantly to improving road safety and emergency response effectiveness in smart transportation environments.

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